TURKU ENERGIA

Tasehinta.

Know when the imbalance price breaks away from spot, and why, before it happens.



Team EDISS #1 · Since AI 2026 · Turku Energia challenge

Every missed plan is settled at a price nobody knows yet.

Finland's imbalance price is set every 15 minutes. It follows spot until the grid runs short of power or into a surplus, then it can jump far either way.

4,000 €/MWh

the imbalance price on 9 Sep at 20:30, with spot at 200

1 in 4

quarter-hours in September ended more than 50 €/MWh from spot

15 min

settlement period: every quarter-hour is its own price

Data: organisers' validation month, 7 Sep – 4 Oct 2026. Photo: Kashima Power Line 07 by Σ64, CC BY 4.0, Wikimedia Commons.



Every hour: the next 40 quarter-hours, four answers each.

What will it cost?

A median imbalance price and an honest 80 % range around it

Which way is the grid going?

The probability of up-regulation, down-regulation or none

Will it spike?

The chance of a price more than 50 €/MWh above or below spot

Why?

The few factors that move the forecast most, strongest first

Then it shows only what deserves attention and says what to check, never what to trade: open data can't see Turku Energia's own position.

One model, built only from what was known at the time.



Published data

Only what was out before the hour: market, balancing and grid data, Fingrid real-time, weather runs, 19 months of history

Features

For each of the 40 quarter-hours: recent state, real-time grid, wind-forecast revisions, the shape of the day-ahead price

One ensemble model

Two gradient-boosting components, averaged: their errors partly cancel

Forecast + dashboard

The submitted forecast file and the situational view

Forecast the gap, not the price. Spot is known the day before; the unknown is how far the imbalance price lands from it.

Direction first. Short of power: above spot. Surplus: below. Price, range, direction and spike risk tell one story.

History made comparable. Pre-June prices rebuilt under the June 2026 rule: 99.5 % exact where it can be checked.

Nothing from the future. Every input used only from its publication time, enforced by automated leak tests.

More accurate than the baseline on every score.

36.7 %

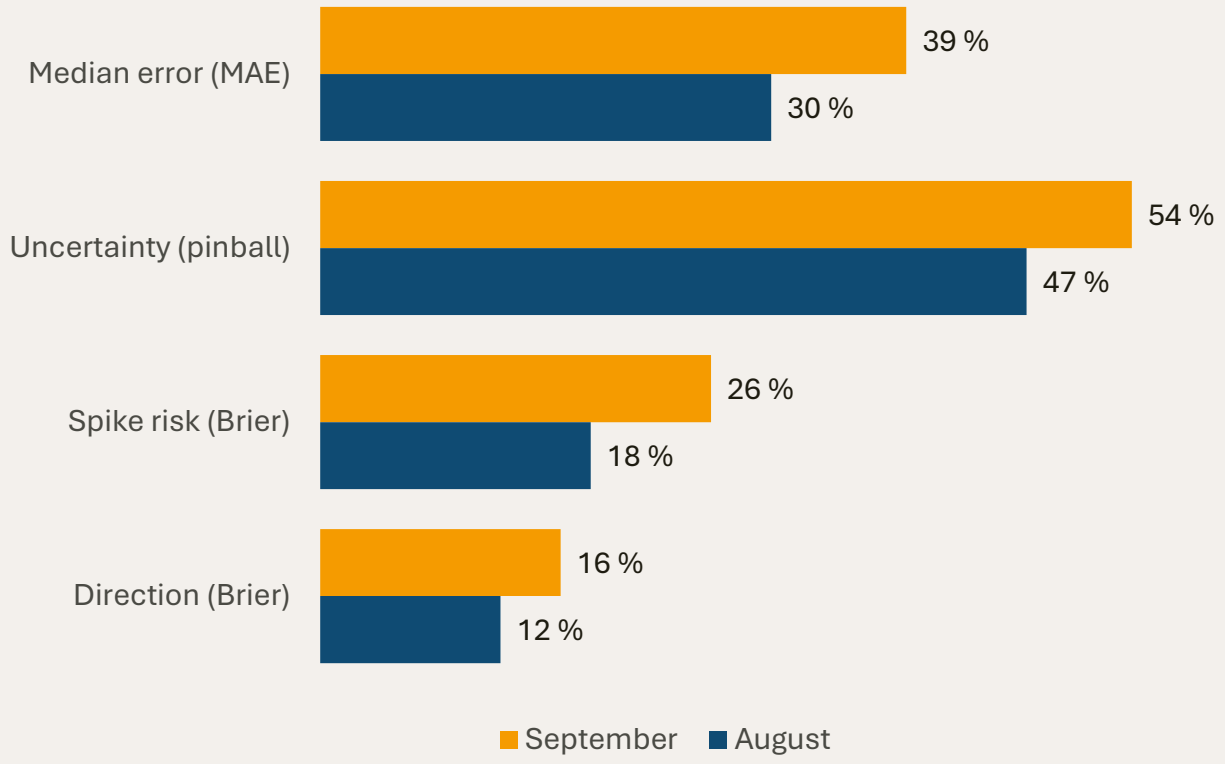
better than the organisers' baseline on their validation month (Sep), weighted by the challenge's points

29.0 %

on August, a second month the model never saw

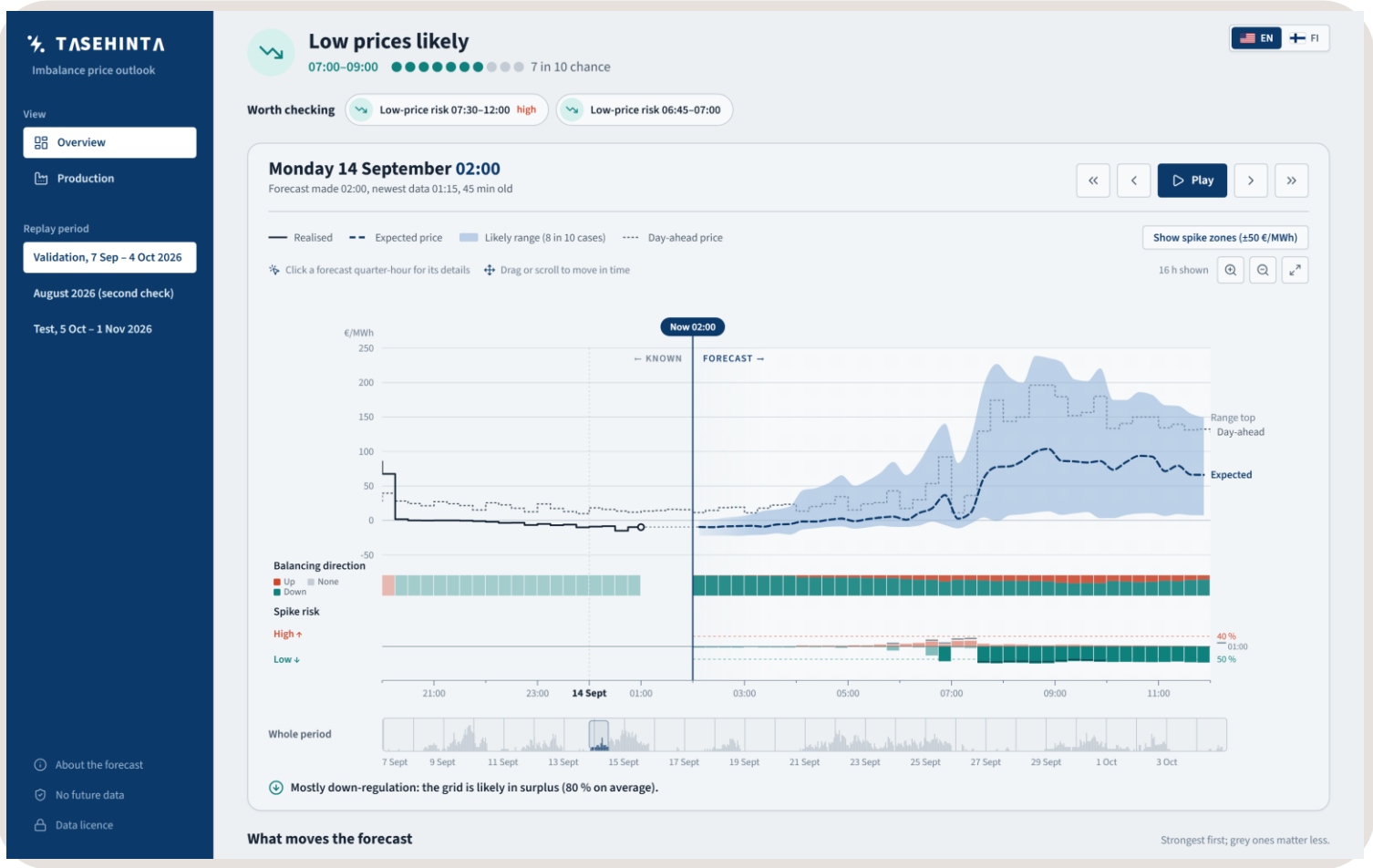
Median error 40.2 vs 66.2 €/MWh · better in every September week · the 80 % range held 81 % of prices

How much smaller our error is than the baseline's



Scored with the organisers' own score.py; each month forecast using only data published before it.

See what's worth checking, and replay any past hour.



Only what matters

Spike risk up, regulation expected, forecast moved, an uncertain signal

One time axis

Price and range, direction and spike risk on one 15-minute grid

Move through time

Scroll back in 15-minute steps; nothing published later is shown

Live, in English and Finnish:
tasehinta.lovable.app

From “check this” to “offer this”.



Turku Energia's own data

Plant capacity, marginal costs and the market position turn a warning into a concrete capacity suggestion.

Winter weather history

The history covers winter 2025–26, but weather-forecast runs only from April 2026; adding winter runs is the next model step.

Live operation

Run the same pipeline every hour on live Fingrid and weather data, behind an API.

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Kiitos. Thank you.

Live: tasehinta.lovable.app

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